Research Poster Manuscript

Discussion Draft v2.0 • Two proposed versions
FoSE UG/PGT Research Showcase 2026
University of Nottingham Ningbo China

我想怎麼安排這張 poster

Poster 預計分成六個部分

1. Research question and motivation：說明 planar localisation 為什麼常用，以及 target 離開 calibration plane 後會產生什麼問題。

2. Geometric model：用 camera、target height 和 projection ray 解釋 off-plane error，保留最關鍵的公式。

3. Experimental method：交代五次 physical rebuilds、兩台 cameras、0–50 mm，以及不使用 nonzero-height observations 擬合預測的 blind validation 設計。

4. Main results：呈現 predicted 與 observed error 的吻合程度，同時保留 geometry 無法解釋的 residual。

5. Engineering interpretation：把幾何誤差、real-system error allowance 和 task tolerance 連起來，討論 planar 何時仍然夠用。

6. Conclusions and limitations：清楚說明研究支持什麼，以及 conditional guarantee、XY metric 和未確認資訊的限制。

我想比較的兩個版本

Version 1 — Task-Conditioned Model Adequacy（我 prefer 這個）
主線是「planar localisation 雖然在幾何上不完全正確，但對指定任務是否仍然足夠？」幾何模型與實驗用來支持最後的 engineering decision。Adequacy map 會是主要視覺重點。

Version 2 — Blind Geometric Prediction & Experimental Validation
主線是「只用 Z = 0 的幾何資訊，能否在不擬合離面資料的情況下，預測真實的 off-plane degradation？」幾何推導與 controlled experiments 是主要內容，task-conditioned adequacy 作為最後的工程延伸。Validation graph 會是主要視覺重點。

兩個版本不是兩組不同的實驗，也不是兩種互相排斥的結論。我們要選的是最適合 showcase 的敘事重點。以下 poster 文案維持英文，中文只用於我們之間的架構說明與比較。

Author(s): [TO BE CONFIRMED] Department: [TO BE CONFIRMED]
Supervisor: [TO BE CONFIRMED]

Version 1

Task-Conditioned Model Adequacy

WHEN IS 3D WORTH IT?

Task-Conditioned Adequacy of Planar Localisation for Vision-Guided Manipulation

Research question: When is planar localisation sufficient, and when is 3D required for a declared task guarantee?

1 Motivation

Planar image-to-workspace mappings simplify calibration and computation in vision-guided manipulation. When a target leaves the calibration plane, its image ray intersects that plane at a displaced XY position, introducing systematic geometric error. A 3D-aware method can improve geometric fidelity, but additional accuracy is useful only relative to task requirements. When does planar localisation remain sufficient, and when is 3D required for a declared task guarantee?

2 Geometric mismatch

Let the camera centre be C = (C_x, C_y, C_z), the target P = (X, Y, h), and the calibration plane Z = 0. Intersecting the camera–target ray with this plane gives the displacement from the true XY position to the planar estimate:

$$\Delta X = h(X - C_x)/(C_z - h)\,;\ \Delta Y = h(Y - C_y)/(C_z - h)$$

$$\rho = \sqrt{[(X - C_x)^2 + (Y - C_y)^2]}\,;\ E_{geom} = |h|\rho/|C_z - h|$$

Error depends jointly on height and lateral camera–target geometry; no universal height threshold determines planar adequacy.

3 Experimental support

Five independent physical rebuilds and two cameras tested target heights from 0 to 50 mm under a fixed-board protocol. Predictions used calibration-plane geometry at Z = 0, without fitting nonzero-height observations.

Validation metric	Reported range
Height-curve correlation	0.9942–0.9951
Point-level magnitude correlation	0.9869–0.9942
Median direction cosine	0.9983–0.9992
Geometry-unexplained vector RMSE	3.61–4.36 mm

These results support prediction of the dominant off-plane magnitude and direction structure. High correlation does not establish complete absolute localisation accuracy; the remaining residual must be retained.

4 Task-conditioned model adequacy

Write the planar XY error as a geometric component plus a real-system residual:

$$e_P = e_{geom} + r_P ; \ \|r_P\| \leq S_P$$

$$E_P = \|e_P\| \leq E_{geom} + S_P$$

For task tolerance τ, the maximum allowable work-plane XY localisation error, the triangle inequality gives a sufficient planar adequacy condition:

$$E_{geom} + S_P \leq \tau$$

A 3D-aware reference is guaranteed within tolerance if S_3D ≤ τ. Only the 3D-aware method provides the declared guarantee when:

$$E_{geom} + S_P > \tau \wedge S_{3D} \leq \tau$$

Failure of the planar sufficient condition does not imply actual planar failure. S_P and S_3D are declared bounded-error scenarios, not universally calibrated constants. Geometric correctness concerns model fidelity; localisation accuracy concerns XY error; task-conditioned adequacy compares an error bound with tolerance. Robust 3D accuracy dominance is a separate sufficient statement:

$$E_{geom} > S_P + S_{3D}$$

Thus, 3D can be more accurate while both methods remain task-adequate.

5 Engineering implications

Engineers can evaluate height, camera placement, workspace position, real-system error allowance and task tolerance together. Under 0 ≤ h < C_z, τ ≥ S_P and ρ + τ − S_P > 0, the planar sufficient condition gives:

$$h_{limit}(\rho) = (\tau - S_P)C_z/(\rho + \tau - S_P)$$

This conditional operating boundary follows from the geometric model; it is neither a new optimisation algorithm nor a universally validated switching threshold. Guarantees concern XY localisation, not complete manipulation success.

6 Conclusions

Predictable geometry: controlled experiments support the projective relationship governing dominant off-plane error.

Task-dependent adequacy: a geometrically imperfect planar model can remain sufficient within the declared tolerance.

Engineering decision: the need for 3D depends on task requirements and declared error bounds.

“3D is not always necessary — even when 2D is geometrically imperfect.”

Version 2

Blind Geometric Prediction & Experimental Validation

PREDICTING OFF-PLANE LOCALISATION ERROR

Blind Geometric Prediction and Physical Validation for Vision-Guided Manipulation

Research question: Can calibration-plane geometry predict off-plane localisation error without fitting observations at nonzero heights?

1 Motivation

Planar image-to-workspace mappings offer simple calibration and inexpensive localisation for robotic manipulation. Their planar assumption is violated when a target rises above the work surface. The resulting error is systematic: the target's viewing ray meets the calibration plane at a displaced position. We investigate whether this degradation can be predicted from geometry before examining the held-out off-plane observations.

2 Analytical prediction

Let C = (C_x, C_y, C_z) be the camera centre and P = (X, Y, h) the target, with calibration plane Z = 0. The planar estimate is the intersection of the camera–target ray with that plane. Its XY displacement is:

$$\Delta X = h(X - C_x)/(C_z - h)\,;\ \Delta Y = h(Y - C_y)/(C_z - h)$$

$$\rho = \sqrt{[(X - C_x)^2 + (Y - C_y)^2]}$$

$$E_{geom} = |h|\rho/|C_z - h|$$

The model predicts both error magnitude and displacement direction from target height and lateral camera–target geometry. Height alone cannot define a universal error threshold. The expression is singular at h = C_z, where ray–plane intersection becomes ill-conditioned.

3 Blind experimental design

Controlled fixed-board experiments used five independent physical rebuilds, two cameras and target heights spanning 0–50 mm. Predictions were obtained from Z = 0 geometry. Nonzero-height observations were held out from fitting the geometric prediction. This design tests whether the analytical mechanism explains real off-plane degradation rather than merely reproducing fitted height trends.

Version 2 Results and engineering interpretation

4 Experimental evidence

Validation metric	Reported range
Height-curve correlation	0.9942–0.9951
Point-level magnitude correlation	0.9869–0.9942
Median direction cosine	0.9983–0.9992
Geometry-unexplained vector RMSE	3.61–4.36 mm

Height-curve and point-level correlations support agreement in the dominant magnitude structure. Direction cosines support the predicted displacement direction. Together, these results validate the dominant geometric behaviour under the tested protocol without fitting held-out nonzero-height observations.

5 What geometry does not explain

The remaining vector RMSE shows that the analytical model does not predict complete real-system accuracy. High correlation is not evidence of small absolute error. The geometry-unexplained residual must remain separate; its causal source is not established, and its RMSE must not be treated as a worst-case error bound.

6 Engineering interpretation

Predictable mismatch can inform task-conditioned adequacy. Let τ be the maximum allowable work-plane XY localisation error. If the non-geometric planar residual satisfies a separately declared bound, the triangle inequality gives:

$$e_P = e_{geom} + r_P\,;\ \|r_P\| \le S_P$$

$$E_P \le E_{geom} + S_P$$

$$E_{geom} + S_P \le \tau$$

The final inequality is sufficient for planar adequacy. A 3D-aware reference is guaranteed within tolerance if S_3D ≤ τ. Only the 3D-aware method provides the declared guarantee when:

$$E_{geom} + S_P > \tau \wedge S_{3D} \le \tau$$

These guarantees depend on declared error scenarios; S_P and S_3D are not universally calibrated constants. Failure to certify planar adequacy does not establish actual failure. Greater 3D accuracy and 3D necessity for a guarantee remain different questions.

7 Conclusions

Prediction: projective geometry explains the dominant off-plane error structure. Validation: controlled physical experiments support blind prediction while retaining a nonzero residual. Application: the value of 3D depends on geometry, system-error allowances and task tolerance. The present evidence concerns XY localisation, not complete manipulation success.

“3D is not always necessary — even when 2D is geometrically imperfect.”

兩個版本的閱讀重點與選擇

Version 1：以工程決策為中心

讀者先理解 planar error 的來源與證據，再把重點放在 $E_{geom} + S_P \le \tau$。最適合強調的結論是：geometric imperfection 不等於 task inadequacy。主要風險是讀者把 declared scenarios 當成已經實驗校準的保證，因此 τ、S_P 和 S_{3D} 的定義要特別清楚。

Version 2：以可驗證的物理預測為中心

讀者先看到模型如何預測，再看 controlled experiments 是否支持預測。最適合強調的是：沒有用 nonzero-height observations 擬合，仍能預測主要的 magnitude 和 direction structure。主要風險是把高 correlation 誤讀為完整 absolute accuracy，因此 residual 結果應與 validation 一起呈現。

Proposed Figures for the Final Poster

Figure specifications and captions below are discussion material, separate from the poster-ready manuscript. No new measured data are proposed.

Figure 1 Off-Plane Geometric Error Model

Purpose: explain the physical origin of planar localisation error. Show camera C, calibration plane Z = 0, target P at height h, the projection ray, true lateral projection Q = (X, Y, 0), and planar estimate P′. Label lateral displacement P′ − Q and its magnitude E_geom. Distinguish this XY displacement from the full 3D distance between P and P′.

Proposed caption: Applying a calibration-plane mapping to an elevated target intersects the viewing ray at P′ rather than Q. The resulting XY error depends on h, ρ and C_z.

Source: manuscript v0.1, §3–4 and Figure 1 (p. 4). Redraw as a schematic; no experimental data required.

Figure 2 Blind Experimental Validation

Purpose: compare analytical prediction and observation across height. Prefer matched predicted and observed height curves, separated by camera with rebuild information retained. If space allows, add point-level predicted-versus-observed magnitudes with a unity line. Define aggregation, error bars, units and sample counts from the frozen analysis; do not reconstruct curves from summary correlations.

Proposed caption: Calibration-plane geometry predicts the dominant off-plane degradation across five rebuilds and two cameras without fitting the nonzero-height observations. Agreement in structure coexists with a nonzero real-system residual.

Source: manuscript v0.1, §5.2, Figure 2 placeholder (p. 5); positioning v1.1, §7.2 and §12.1. Use frozen Stage 2/E2 outputs only. Exact plot file, data table, commit, aggregation and per-camera metric assignment: [TO BE CONFIRMED]. The supplied PDFs contain a placeholder, not the final validation graph.

Figure 3 Task-Conditioned Adequacy Map

Purpose: show conditional planar sufficiency. Use target height h on the horizontal axis and lateral distance ρ on the vertical axis. Plot E_geom + S_P = τ for explicitly declared C_z, S_P and τ; shade the region satisfying the inequality. Label the other region “planar not guaranteed”, not “planar failure”.

Evaluate S_3D ≤ τ separately. Only when this condition holds may the non-certified planar region be labelled “3D-only guaranteed under declared scenarios”. If it fails, that region is “neither guaranteed”. This is an analytical scenario map, not measured task-success data.

Proposed caption: Task tolerance and declared system-error allowances determine planar sufficiency within the stated geometry. The 3D-aware guarantee is evaluated independently.

Source: manuscript v0.1, §6 and Figure 4 (p. 6); positioning v1.1, §6.3–6.5. Scenario values and a defensible task-specific τ: [TO BE CONFIRMED].

Derivation Notes

Ray–plane intersection

Assume a valid central-projection ray in a common calibrated coordinate frame. Let R(t) = C + t(P − C). Solving its Z coordinate for the plane gives:

$$0 = C_z + t(h - C_z) \Rightarrow t = C_z/(C_z - h), \qquad h \neq C_z$$

$$P'_{XY} - P_{XY} = (t - 1)(P_{XY} - C_{XY})$$

$$e_{geom} = h(P_{XY} - C_{XY})/(C_z - h)$$

Taking the Euclidean norm yields E_geom = |h|ρ/|C_z − h|. For 0 < h < C_z, the displacement points away from the camera's lateral projection. At h = C_z, the plane-intersection formula is singular; near this height it is ill-conditioned.

Adequacy and accuracy dominance

$$max(0, E_{geom} - S_P) \leq E_P \leq E_{geom} + S_P$$

The upper bound follows from the triangle inequality; the lower bound follows from the reverse triangle inequality. If E_geom > S_P + S_3D, the planar lower bound exceeds the 3D-aware upper bound. This establishes robust error dominance under the declared scenarios, independently of τ.

Operating boundary

Let T = τ − S_P ≥ 0 and C_z > 0. For 0 ≤ h < C_z:

$$h\rho/(C_z - h) \leq T \quad \Leftrightarrow \quad h(\rho + T) \leq TC_z$$

Division by ρ + T > 0 gives h ≤ h_limit(ρ), intersected with the original domain h < C_z. If ρ = 0 and T > 0, h_limit = C_z but the singular endpoint is excluded. If ρ = T = 0, the boundary formula is undefined; E_geom = 0 throughout the nonsingular domain. If τ < S_P, this sufficient bound cannot certify any point.

Evidence traceability and open items

The four validation ranges and physical configuration are reported in manuscript v0.1, Abstract (p. 2), §5 (p. 5), and positioning v1.1, §7.2–7.3 (p. 8). They are reported summary results, not recomputed estimates. The residual RMSE is neither a worst-case bound nor an estimate of S_P.

Before submission: exact hardware and C_z; height list; target coordinates; sample/frame counts; calibration and ground-truth procedures; exclusions; frozen data/plot paths; and task-derived τ: [TO BE CONFIRMED]. Retain sensitivity labels until the declared bounds can be defended for the intended operating domain.

Scientific references and positioning

The integrated contribution combines blind physical validation of projective mismatch with separate system-error scenarios and task-conditioned guarantee logic. Prior work already links planar-assumption error to task requirements; no first-ever claim is made [1–3]. References below are reproduced from manuscript v0.1, References (p. 10).

[1] J. Carlson, "Flatland Revisited: The Effects of Non-Planar Floors on Large-Scale Warehouse Deployment of Autonomous Mobile Robots," ICRA 2025 Workshop on Robots in the Wild, 2025.

[2] J. Carlson and A. Kolling, "Selective Placement of Visual Fiducials for Localization Tasks," ICRA Workshop on Robot Execution Failures and Failure Management Strategies, 2023.

[3] S. Liu, X. Huang, N. Fu, C. Li, Z. Su, and S. Ostadabbas, "Simultaneously-Collected Multimodal Lying Pose Dataset: Enabling In-Bed Human Pose Monitoring," IEEE TPAMI, 45(1), 1106–1118, 2023. doi: 10.1109/TPAMI.2022.3155712.

Discussion Points for Research Partners

Partner 1: _____ Partner 2: _____
Discussion date: _____

Topic	Question to Discuss	Decision / Notes
1. Version choice	Which central story is clearer for the showcase: Version 1 Task-Conditioned Model Adequacy, or Version 2 Blind Geometric Prediction & Experimental Validation?	Partner 1: Partner 2: Decision:
2. Evidence strength	Does the geometric validation provide sufficient evidence for the intended claims?	Partner 1: Partner 2: Decision:
3. Residual results	Should we include the geometry-unexplained residual results in the poster?	Partner 1: Partner 2: Decision:
4. Blind prediction graph	Which experimental graph best demonstrates the blind prediction?	Partner 1: Partner 2: Decision:
5. Operating envelope	Should we include the operating-envelope equation?	Partner 1: Partner 2: Decision:
6. Task tolerance	Do we have a defensible real manipulation-task tolerance τ, or should all examples remain clearly labelled sensitivity scenarios?	Partner 1: Partner 2: Decision:
7. Declared bounds	Are S_P and S_3D adequately defined and clearly presented as assumptions?	Partner 1: Partner 2: Decision:
8. Closest prior work	Are the claims sufficiently distinct from the closest prior work?	Partner 1: Partner 2: Decision:
9. Figure priorities	Which three figures should receive the most poster space?	Partner 1: Partner 2: Decision:
10. Experimental scope	Does the poster accurately reflect the experiments we actually completed?	Partner 1: Partner 2: Decision:

Agreed revisions and next actions: _____
_____